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Statistics in Biopharmaceutical Research

ISSN: 1946-6315 (Online) Journal homepage: www.tandfonline.com/journals/usbr20

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STATISTICS IN BIOPHARMACEUTICAL RESEARCH
2025, VOL. 00, NO. 0, 1–8
https://doi.org/10.1080/19466315.2025.2539831

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A Multi-Arm Multi-Stage Group Sequential Phase 2/3 Design with Dose Selection for Oncology Trials

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ABSTRACT
Seamless phase 2/3 design with dose selection has generated increasing interest in clinical trial design, aiming to expedite drug development with fewer participants. The existing designs consist of two stages, with multiple doses evaluated in the first stage (phase 2) and the selected dose (s) tested in the second stage (phase 3). In this article, we extend the two-stage design to a multi-stage design that selects a single dose and tests it based on a time-to-event endpoint. The proposed design controls the Type-I error rate by using the combination test along with the group sequential method and the closed testing procedure. The simulation results confirm the control of Type-I error rate and indicate that the proposed design offers attractive operating characteristics compared to the traditional sequential approach with separate phase 2 and 3 trials.

ARTICLE HISTORY
Received October 2024
Accepted July 2025

KEYWORDS
Adaptive design; Dose selection; Group sequential; Multi-arm multi-stage (MAMS); Seamless phase 2/3 trial; Survival endpoint

1. Introduction

In many therapeutic areas, the typical drug development approach involves conducting randomized phase 2 trials (e.g., dose-ranging study) to select the optimal dose, followed by confirmatory phase 3 trials in a sequential manner. In oncology, dose selection is primarily conducted during phase 1 dose-finding trials, with the dose-ranging trials often being skipped. Typically, the maximum tolerated dose (MTD) is chosen for the phase 3 trial. With the advent of new anti-cancer therapies, a paradigm shift is underway from the use of the conventional MTD approach to improved dose selection strategies for oncology programs. To encourage dose optimization in oncology drug development, the FDA's Oncology Center of Excellence launched the project Optimus in 2021 and, in 2023, published the guidance "Optimizing the Dosage of Human Prescription Drugs and Biological Products for the Treatment of Oncologic Diseases" (U.S. Food and Drug Administration 2023). Efforts toward dose optimization prior to approval will increase the likelihood of delivering the optimal dose to patients. However, this approach requires greater investment from sponsors and presents challenges to ensure rapid access to safe and effective new therapies.

The adaptive seamless phase 2/3 designs, which allow for dose selection in the phase 2 portion and confirmatory testing in the phase 3 portion have gained increasing interest in the recent years. The adaptive design minimizes the time gap between phase 2 and phase 3 trials that would occur if they were conducted separately. It also increases data efficiency by integrating data from both phases into the statistical testing. With the addition of the mid-trial adaption for dose selection, it is essential to correct for the multiplicity that arises from comparing multiple dose arms to a common control. In oncology trials, the primary efficacy endpoint, overall survival (OS), is often immature at the time of dose selection. As a result, early surrogate endpoints such as objective response rate (ORR) or progression-free survival (PFS) are commonly used to inform adaptive decisions. Even though OS is not used for dose selection, inflation of Type I error could occur if aggregated OS data is directly used at the final analysis in phase 3, due to the positive correlation between the surrogate endpoint used for dose selection and OS. Some adaptive designs address the multiplicity issue by using Bonferroni correction. For example, the 2-in-1 designs allow a trial to conclude as either a phase 3 or phase 2 trial. Those designs allow the use of the conventional log-rank test based on cumulative OS data by assuming the correlation between the surrogate endpoint used for dose selection and OS is stronger at the end of a phase 2 trial than at the end of a phase 3 trial (Jin and Zhang 2021; Zhang et al. 2022, 2023). The graphical approach based on weighted Bonferroni tests is used to control the Type I error rate. The Bonferroni correction based on cumulative data is conservative as it imposes an excessive penalty on the incremental data collected after dose selection. Some adaptive designs employ more complex multiplicity adjustments to use data more efficiently. In a two-stage design, OS data collected before and after dose selection can be combined using a combination test (Bauer and Köhne 1994; Lehmacher and Wassmer 1999). Some authors have proposed using adjusted p-values based on stage-1 OS data (Li et al. 2015; Wang et al. 2023), while others have recommended using a general closed testing procedure to control the Type-I error rate (Bauer and Kieser 1999; Posch et al. 2005; Friede et al. 2011). For time-to-event

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endpoint, (Bauer and Posch (2004)) pointed out that the Type-I error rate could be inflated if the dose selection is affected by the early surrogate endpoint from stage-1 patients and OS follow-up of these stage-1 patients contributes to the stage-2 log-rank test. One solution is to apply a combination test to OS data from cohorts of patients enrolled before and after the dose selection (Friede et al. 2011; Jenkins, Stone and Jennison 2011).

In this article, we propose a phase 2/3 adaptive design that selects a single dose for phase 3 and extends the existing two-stage design to multiple stages, allowing interim analysis, with the intention to stop early for efficacy, in the phase 3 portion. The inverse normal combination test (Lehmacher and Wassmer 1999) is used to combine data from patients enrolled before and after dose selection. The closed testing procedure (Bauer and Köhne 1994) is employed to adjust for testing multiple dose arms against a common control arm. The classical group sequential test is used to adjust for multiplicity arising from testing a single hypothesis across multiple stages.

The remainder of the article is organized as follows. In Section 2, we present the proposed design and discuss the control of Type I error. In Section 3, we illustrate the design using a hypothetical example. In Section 4, we conduct simulation studies to confirm the Type I error rate and evaluate the operating characteristics of the proposed design. We conclude the article with a discussion in Section 5.

2. Materials and Methods

2.1. The Proposed Design

Consider a trial with K dose candidates of an investigational drug. The primary endpoint is OS. The proposed design will first enroll n_1 patients per arm in a phase 2 trial starting with K doses of the investigational drug and one control arm. An interim analysis (stage 1) in phase 2 will be performed for dose selection. At stage 1, each of the dose arms will be compared to the control arm based on an intermediate or surrogate efficacy endpoint (e.g., ORR of PFS). In addition to the efficacy data, the totality of data including pharmacokinetics (PK) and safety, is also typically assessed. After stage 1, at most one dose will be selected based on the overall benefit-risk profile. If there is no dose arm demonstrates a promising benefit-risk profile compared to the control arm, the trial will remain in a phase 2 study, with no intention for formal statistical testing. Note that there is no formal statistical testing, nor intention to stop early for efficacy at stage 1.

If the trial is expanded to a phase 3 trial, additional n_2 patients per arm will be enrolled to the selected dose arm and the control arm. Similar to a traditional group sequential phase 3 trial, we can test the primary endpoint of OS at interim analyses (IA), starting at the end of stage 2, with the intention to stop early for efficacy before the final analysis (FA). For example, a 3-stage design will include stage 1 for dose selection, with stage 2 and 3 serving as IA and FA, respectively, for testing OS of the selected dose. The study design is summarized in Figure 1 using a hypothetical example with three dose arms and three stages. Since multiple dose arms are compared to the control arm at stage 1, multiplicity adjustment is needed due to the correlation between the surrogate endpoint used for dose selection and OS. In addition, proper adjustment is needed for multiplicity induced by testing the selected dose multiple times across stages in phase 3. Note that patients in the unselected dose arms need to be followed for OS until the end of the study, as their OS data are required for statistical testing. We describe the methods for controlling the Type I error rate in Section 2.2.

2.2. Control of Type I Error

Consider a trial comparing K dose arms, indexed by $i = 1, 2, \ldots K$, to a common control arm. Let H_0^i denote the null hypothesis of no difference in OS between dose arm i and control arm. Let $j = 1, 2, \ldots S$ denote the stages where $S \geq 2$ and $j = 1$ for dose selection. For simplicity, we assume same number of patients will be enrolled for each arm. Specifically,

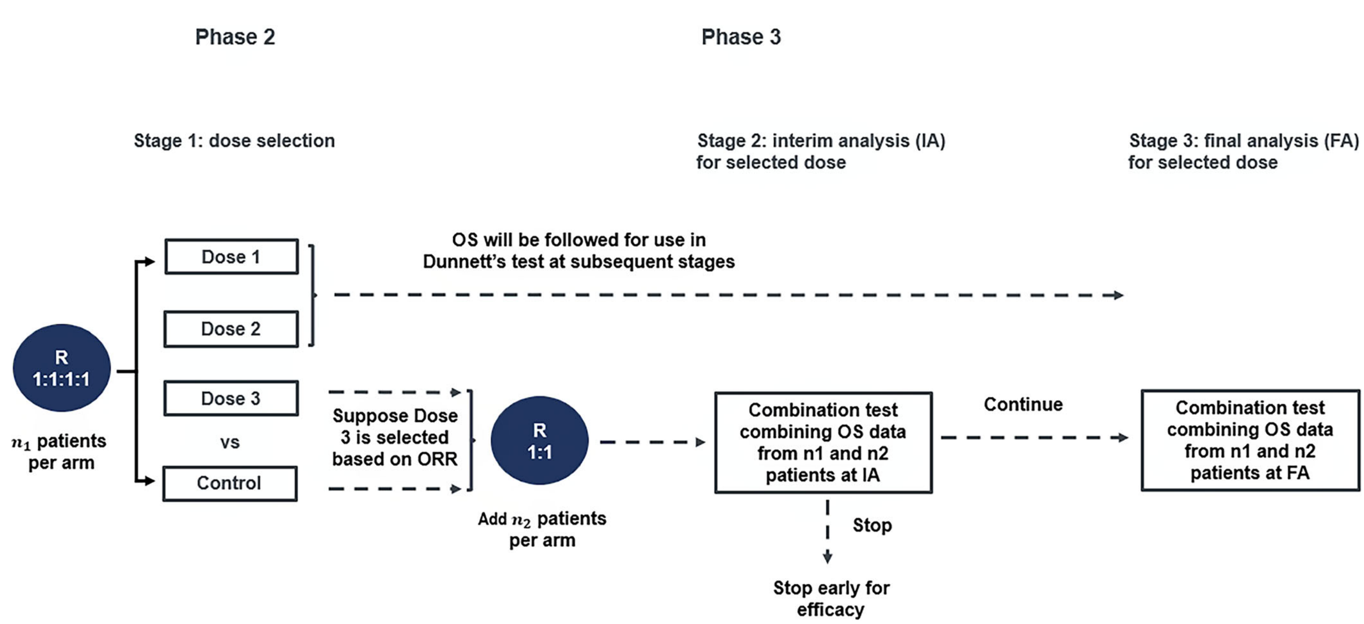

Figure 1. Flowchart of the proposed design illustrated with a hypothetical example with three dose arms and three stages.

n_1 patients per arm will be enrolled before dose selection (stage 1). After dose selection, n_2 patients per arm will be added to the selected dose arm and the control arm.

We use the closed testing procedure (Bauer and Köhne 1994) together with the inverse normal combination test (Lehmacher and Wassmer 1999) to adjust for the multiplicity arising from testing multiple dose arms to a common control. The individual null hypothesis H_0^r associated with the selected dose arm r is only rejected overall if, all intersection hypotheses $H_0^I = \cap_{g\in I} H_0^g$ where $I \subseteq \{1, \dots, K\}$ with $r \in I$, are rejected at local level α (1-sided). At each stage after stage 1, the log-rank test statistics are generated separately for patients enrolled before and after stage 1 dose selection. Each intersection hypothesis H_0^I then is tested based on the two sets of statistics, combined using the inverse normal combination method.

For the first set of statistics based on the cohort of patients enrolled before stage 1, the inverse normal combination method is also used to combine OS data across multiple stages. For each H_0^I, let $Z_{1(j)}^i$ denote the standardized incremental log-rank test statistic based on OS data accumulated from stage $j-1$ to stage j for $j \geq 2$. Note that $Z_{1(2)}^i$ is equal to the standardized cumulative log-rank test statistic up to stage 2 as there is no statistical testing at stage 1. $Z_{1(j)}^i, j = 2, \dots, S$, are independent and each follows a standard normal distribution under H_0^i (Tsiatis 1981). Define $Z_{1(j)}^I = \left\{Z_{1(j)}^i, i \in I\right\}$. $Z_{1(j)}^I$ asymptotically follow a multivariate normal distribution with $E\left(Z_{1(j)}^i\right) = 0$, $var\left(Z_{1(j)}^i\right) = 1$, $cov\left(Z_{1(j)}^i, Z_{1(j)}^{i'}\right) = 1/2$ for $i \neq i'$ under H_0^I (Dunnett and Tamhane 1992). Let $Z_{1(j)}^{I_{\max}} = \max_{i\in I}\left\{Z_{1(j)}^i\right\}$. The corresponding Dunnett p-value based on the incremental statistics is $p_{1(j)}^I = P\left(Z_{1(j)}^{I_{\max}} > \max_{i\in I}\left\{z_{1(j)}^i\right\}\right)$. $p_{1(j)}^I$ follows a uniform distribution under H_0^I. The inverse normal test statistic for H_0^I based on n_1 patients per arm at stage $j \geq 2$ is calculated as

$$Z_{1j}^I = \sum_{l=2}^{j} w_{jl} \Phi^{-1}\left(1 - p_{1(l)}^I\right), \tag{1}$$

where $\Phi()$ is the cumulative distribution function of the standard normal distribution, and w_{jl} are pre-specified weights with $0 < w_{jl} < 1$ and $\sum W_{jl}^2 = 1$. A commonly used option for the weights is to make w_{jl} proportional to the square roots of the number of events expected at each stage. Since $\Phi^(-1)(1-p_{1(j)}^I)$'s are standard normally distributed, Z_{1j}^I follows a standard normal distribution under H_0^I.

For the second set of statistics based on the cohort of patients enrolled after stage 1, the cumulative log-rank test statistics for the selected H_0^r will be used for each H_0^I because the additional n_2 patients per arm are enrolled only for the selected dose arm and the control arm. Let Z_2j^r denote the standardized log-rank test statistic at stage $j = 2$ for testing H_0^r, where r is the selected dose arm. Let Z_2j^I denote the test statistic based on the n_2 patients per arm for H_0^I. $Z_{2j}^I = Z_{2j}^r$ and follows a standard normal distribution under H_0^I.

The inverse normal test statistic combing the two cohorts of patients for H_0^I at stage $j = 2$ is calculated as

$$Z_j^I = h_1 j Z_1 j^I + h_2 j Z_2 j^I, \tag{2}$$

where h_{1j} and h_{2j} are pre-specified weights with $h_{1j}^2 + h_{2j}^2 = 1$. A commonly used option for the weights is to make them proportional to the square roots of the number of events or sample size planned from respective cohort of patients at each stage. Note that we set the weights to be same for all possible H_0^I. One can also pre-specify the weights differently for each H_0^I if needed. Under H_0^I, $Z_j^I, j = 2, \dots, S$, follow a multivariate normal distribution with E $(Z_j^I) = 0$, var $(Z_j^I) = 1$. The covariance between Z_j^I and $Z(j')^I$ for $j < j'$ can be calculated as

$$\begin{aligned} cov(Z_j^I, Z(j')^I) &= h_1 j h_{1j'} cov\left(Z_{1j}^I, Z_{1j'}^I\right) + h_{2j} h_{2j'} cov\left(Z_{2j}^I, Z_{2j'}^I\right) \\ &= h_{1j} h_{1j'} \sum_{l=2}^{j} w_{jl} w_{j'l} + h_{2j} h_{2j'} \sqrt{d_{2j}/d_{2j'}} \end{aligned} \tag{3}$$

where $d_{2j}, j = 2, \dots, S-1$, are the actual number of OS events from the second cohort of patients observed at each stage.

A multi-stage level-α test of H_0^I can be constructed as follows. Suppose $\alpha_j, j = 2, \dots, S$ are the alpha spent at each stage based on a certain α spending function. For any H_0^I with $r \in I$, H_0^I is rejected at stage $j \geq 2$ if $Z_j^I \geq c_j$ where c_j satisfy the following probability equation.

$$P\left(\left\{Z_j^I \geq c_j\right\} \cap_{j'=2}^{j-1} \left\{Z_{j'}^I < c_{j'}\right\}\right) = \alpha_j, \tag{4}$$

Since $Z_j^I, j = 2, \dots, S$, follow a multivariate normal distribution with the known covariance matrix based on (3), we can obtain the efficacy boundary c_j using the standard group sequential methods. Note that if a H_0^I is rejected at a certain stage $j \geq 2$, we consider it is rejected for all subsequent stages. The total Type-I error rate for H_0^I is maintained at $\sum_{j=2}^{S} \alpha_j = \alpha$. We reject H_0^r for the selected dose arm r with strong control of Type-I error rate of α if all H_0^I with $r \in I$ are rejected at level-α. The testing procedure can be summarized in six steps:

1. Divide patient population into two cohorts. Cohort 1 includes patients enrolled before dose selection (stage 1) across all arms. Cohort 2 includes patients enrolled after dose selection for the selected dose and the control arm.
2. At stage $j \geq 2$, for each intersection hypothesis, obtain the Dunnett p-value based on the incremental statistic accrued from the previous stage of the primary endpoint for cohort 1. Obtain the standard cumulative log-rank test statistic up to stage j for the selected dose based on cohort 2.
3. For cohort 1, for each intersection hypothesis, combine the Dunnett p-values across previous stages, starting from stage 2, using the inverse normal combination test based on (1). Note that at stage 2, this step can be skipped, as the Dunnett p-values from Step-2 are directly based on cumulative statistics.
4. For each intersection hypothesis, combine the Dunnett p-value for cohort 1 and standard log-rank test p-value for cohort 2 using the inverse normal combination test based on (2).

5. For each intersection hypothesis, calculate the efficacy boundaries using the standard group-sequential method. If the same combination test weights are applied for each intersection hypothesis, the boundaries will be same for all intersection hypotheses.
6. According to the closed testing procedure, H_0^r for the selected dose arm r is rejected only if all H_0^I with $r \in I$ are rejected at level-α.

3. Illustrated Example

Consider a trial starts with 3 dose arms along with a common control arm. Stage 1 for dose selection occurs when $n_1 = 60$ patients per arm are enrolled for each arm. The sponsor plans to select a dose arm with the best ORR that shows at least 10% ORR difference compared to the control. Based on the observed ORR data at stage 1 (Table 1), dose 3 is selected into phase 3. The primary endpoint is OS and let H_0^i denote the null hypothesis of no difference in OS between dose arm i and control arm. We reject H_0^3 when all intersection hypotheses containing H_0^3 are rejected at level $\alpha = 0.025$. This produces four hypotheses: $H_0^3, H_0^{13} = H_0^1 H_0^3, H_0^{23} = H_0^2 H_0^3, H_0^{123} = H_0^1 H_0^2 H_0^3$.

An additional $n_2 = 240$ patients per arm are enrolled for dose 3 and control arm. As a result, a total of 600 patients will be enrolled in dose 3 and the control arm. Two stages are planned in phase 3 with stage 2 as an efficacy interim analysis and stage 3 as the final analysis. Suppose the sponsor plans to trigger stage 2 when 388 OS events have occurred, and to trigger stage 3 when 473 OS events have occurred among the 600 patients. At each stage, OS data is first analyzed separately for patients enrolled before (cohort 1) and after (cohort 2) stage 1, and then combined using the inverse normal combination test. This follows Step 1 of the testing procedure outlined at the end of Section 2.

For each of the intersection hypothesis, following Step 2 of the testing procedure outlined at the end of Section 2, the OS data in cohort 1 is summarized using Dunnett's test, including all relevant dose arms in that intersection hypothesis. Note that for patients in the unselected dose arms (dose 1 and dose 2), they need to be followed up for OS after stage 1, and their OS data at stage 2 will be used in Dunnett's test. The OS data in cohort 2 is the same for all the intersection hypotheses and is summarized using the log-rank test comparing dose 3 versus control. Note that at stage 2, we can skip the step of combining OS data across stages for cohort 1 (Step 3 of the testing procedure outlined at the end of Section 2), as stage 2 is the first stage for statistical testing. The weights to combine the two cohorts are pre-specified as $h_{12} = \sqrt{95/388}$ and $h_{22} = \sqrt{293/388}$ for cohort 1 and cohort 2, respectively, where 95 is the expected number of events to occur from cohort 1 at stage 2. The inverse normal combination test is then constructed based on (2) in Section 2.2 following Step 4 of the testing procedure outlined at the end of Section 2. Next, we calculate the efficacy boundaries using the group sequential method per Step 5 in the testing procedure outlined at end of Section 2. The p-value boundary is calculated using a Lan-DeMets spending function approximating O'Brien-Fleming bounds. An $\alpha_2 = 0.0133$ is spent at stage 2, based on the information fraction calculated as 388/473. A level-α_2 combination test is constructed for each of $H_0^3, H_0^{13}, H_0^{23}, H_0^{123}$. The null hypotheses are tested per Step 6 in the testing procedure outlined at end of Section 2. The results are summarized in Table 2. As a result, H_0^{23}, H_0^{123} are rejected at stage 2 and the trial continues to stage 3 for H_0^3, H_0^{13}.

At stage 3, Steps 1 and 2 of the testing procedure are repeated using data at stage 3. In addition, OS data from cohort 1 are combined across two stages using (1) in Section 2.2, as described in Step 3 of the testing procedure outlined at end of Section 2. 107 events are expected to occur from cohort 1 at stage 3. The combination test weights are pre-specified as $w_{32} = \sqrt{95/107}$ and $w_{33} = \sqrt{12/107}$ for stage 2 and stage 3, respectively. The results are summarized in Table 3. We then combine the two cohorts per Step 4 of the testing procedure outlined at end of Section 2, The weights to combine the two cohorts are pre-specified as $h_{13} = \sqrt{107/473}$ and $h_{23} = \sqrt{366/473}$ for cohort 1 and cohort 2, respectively. The actual number of events are used to calculate the efficacy boundaries per Step 5 of the testing procedure outlined at end of Section 2. The actual number of events from cohort 2 are $d_{22} = 298$ and $d_{23} = 371$ at stages 2 and 3, respectively. The p-value boundary is 0.021, obtained using (4) in Section 2.2 with the correlation calculated from (3) in Section 2.2. Lastly, we test the remaining null hypotheses per Step 6 of the testing procedure. For each of H_0^3, H_0^{13}, we reject the null hypothesis if the corresponding combination test p-value is less than 0.021. The results are summarized in Table 4. Both H_0^3 and H_0^{13} are rejected at stage 3. As a result, H_0^3 is rejected overall and the sponsor concludes that dose 3 of the investigational drug

Table 1. Observed ORR at stage 1.

	Dose 1	Dose 2	Dose 3	Control
ORR	18%	25%	33%	11%
ΔORR*	7%	14%	22%	–

*ΔORR is the ORR difference between a dose arm and the control arm.

Table 2. Results of the combination test combing OS data from two cohorts at stage 2.

	Dunnett's test based on cohort 1*	Log-rank test based on cohort 2*	Combination test
H_0^3			
p-value (1-sided)	0.658	0.008	0.030
H_0^{13}			
p-value (1-sided)	0.701	0.008	0.033
H_0^{23}			
p-value (1-sided)	0.325	0.008	0.010
H_0^{123}			
p-value (1-sided)	0.370	0.008	0.012

*Cohorts 1 and 2 refer to patients enrolled before and after stage 1 dose selection, respectively. Each hypothesis is rejected at a significance level of $\alpha = 0.0133$.

Table 3. Results of combination test combing two stages for cohort 1 at stage 3.

	Dunnett's test at stage 2	Dunnett's test at stage 3	Combination test
H_0^3			
p-value (1-sided)	0.658	0.061	0.446
H_0^{13}			
p-value (1-sided)	0.701	0.106	0.531

Table 4. Results of the combination test combing OS data from two cohorts at stage 3.

	Dunnett's test based on cohort 1*	Log-rank test based on cohort 2*	Combination test
H_0^3			
p-value (1-sided)	0.446	0.007	0.013
H_0^{13}			
p-value (1-sided)	0.531	0.007	0.017

*Cohorts 1 and 2 refer to patients enrolled before and after stage 1 dose selection, respectively. Each hypothesis is rejected at a significance level of $\alpha = 0.021$.

demonstrates statistically significant better OS compared to the control drug.

4. Simulation Studies

In this section, we conduct simulations to confirm the Type I error control, as well as to evaluate the operating characteristics of the proposed design. Similar to the illustrated example in Section 3, we consider a trial starts with 3 dose arms along with a common control arm. OS is the primary endpoint and ORR is assessed at stage 1 for dose selection. For simplicity, the dose arm with the best ORR compared to control is selected for phase 3. Stage 1 for dose selection is planned to occur after enrolling n_1 patients per arm. Following stage 1, one dose will be selected, and an additional n_2 patients per arm will be enrolled for the selected dose and the control arms. The total sample size $(n_1 + n_2)$ for the selected dose and the control arms is fixed at 600 patients. Two stages are planned in phase 3 for statistical testing of OS. Specifically, stage 2 is planned to occur when 388 OS events have been realized, and stage 3 is scheduled upon reaching 473 OS events.

Subject-level ORR and OS data are generated with a positive correlation ρ. Specifically, OS time-to-event data is generated by an exponential distribution with a specified median OS (mOS). Each patient's time to death is then divided into one-month intervals, and a response status (1 for response, 0 for no response) is generated based on a Bernoulli distribution with a specified probability within each interval. A patient is declared a responder if at least one interval records a response of 1. By varying the probability within each interval, we can generate ORR data that meet an assumed response rate and exhibit a defined correlation with OS. We assume the control arm has a mOS of 15 months.

4.1. Type I Error Evaluation

For Type I error evaluation, the mOS across dose arms is set to be the same as it in the control arm. We vary the values of n_1, ρ and ORRs to assess whether the Type I error rate can be controlled across different scenarios. We consider four scenarios. In scenarios 1–3, there are no differences in ORR between any of the dose arm and the control arm. The ORR is set at 10% with a correlation of 0.3 to OS in scenario 1. In scenarios 2 and 3, the ORRs are set at 20% and 35%, respectively, with correlations to OS set at 0.4 and 0.5. In scenario 4, the ORR in the control arm is set at 35%, and we assume that each dose arm demonstrates an increase of 10% in ORR compared to the control. The correlation between ORR and OS is set at 0.6. In each scenario, we vary the value of n_1 from 30 to 210.

The settings for each scenario and the estimates for the Type I errors are summarized in Table 5. The Type I error rate is controlled at less than 2.5% across all scenarios evaluated. The testing procedure is often conservative, with the degree of conservatism depending on ρ and n_1. Given the same value of n_1, the approach becomes increasingly conservative as the correlation decreases. This is because the treatment selection is based on the early endpoint ORR, while the multiplicity adjustment via the Dunnett's test implicitly assumes that the selected treatment has the best outcome in the primary endpoint OS. When correlation is fixed, the approach becomes increasingly conservative as n_1 increases. This is because larger multiplicity adjustment is made through cohort 1 when more patients are enrolled for dose selection. The Type I error rate can also be controlled when ORR is not under the null hypothesis and is positively correlated with the OS.

Four values in the first column represent the ORR in dose 1, dose 2, dose 3 and the control arm, respectively. ρ is the correlation between ORR and OS. n_1 is the sample size per arm at stage 1 for dose selection. Each row is based on 50,000 simulations of a trial under the global null hypothesis of OS where mOS are same across arms.

4.2. Power Evaluation

For power exploration, we consider dose 1 as the optimal dose and vary its ORR and mOS across different scenarios. The ORR is set at 20% for Doses 2 and 3 and at 10% for the control arm. The mOS is set at 18 months for Doses 2 and 3 and at 15 months for the control arm. For dose selection, in addition to selecting the winner based on ORR, we require the chosen dose arm to demonstrate a ΔORR greater than 10% compared to the control arm.

The first question we aim to address is whether the seamless Phase 2/Phase 3 adaptive design offers an advantage compared to the traditional drug development strategy of conducting separate phase 2 and phase 3 trials sequentially, assuming the sample size and follow-up durations are identical. We consider the sequential approach starts with a randomized phase 2 trial for dose selection. The sample size, timing of the dose selection and selection criteria are identical to those used in stage 1 of the proposed adaptive design. Subsequently, a separate phase 3 group sequential trial is conducted with an efficacy IA followed by a FA to test OS for the selected dose. The sample size and the lengths of follow-up for IA and FA are set to be identical to those used in stages 2 and 3 of the proposed adaptive design. As a result, number of OS events at IA and FA in the separate phase 3 trial will be smaller than the numbers in stages 2 and 3 of the proposed design, as the adaptive design also includes events from the patients enrolled for dose selection. We label the proposed adaptive design as "Seamless Phase2/Phase3," and the sequential approach as "Sequential Phase2/Phase3." For the proposed adaptive design, power is defined as the percentage of simulated replicates that select dose 1 and reject the null hypothesis for OS in the phase 3 portion. In contrast, for the sequential approach, power is defined as the percent-

Table 5. Estimated Type I error rates under global null hypothesis of OS.

ORR	ρ	$n_1 = 30$	$n_1 = 60$	$n_1 = 90$	$n_1 = 120$	$n_1 = 150$	$n_1 = 180$	$n_1 = 210$
0.1, 0.1, 0.1, 0.1	0.3	0.0217	0.0197	0.0188	0.0189	0.0178	0.0171	0.0154
0.2, 0.2, 0.2, 0.2	0.4	0.0202	0.0210	0.0210	0.0197	0.0186	0.0175	0.0175
0.35, 0.35, 0.35, 0.35	0.5	0.0225	0.0216	0.0212	0.0201	0.0205	0.0202	0.0194
0.45, 0.45, 0.45, 0.35	0.5	0.0239	0.0228	0.0219	0.0222	0.0213	0.0207	0.0200

age of simulated replicates that select dose 1 from a phase 2 trial and reject the null hypothesis for OS in a separate phase 3 trial.

In Table 6, we consider four scenarios where the ORR of dose 1, the optimal dose, increases from 0.25 to 0.4. The mOS remains the same across all four scenarios. In each scenario, we further vary n_1 from 30 to 120. In all scenarios, the probability of selecting the optimal dose increases as the sample size for dose selection (n_1) grows. The power for Seamless Phase2/Phase3 also consistently increases as n_1 grows. The results suggest that the higher chance of selecting the optimal dose outweighs the larger α penalty incurred for dose selection, both due to the larger sample size n_1. The power for Sequential Phase2/Phase3 initially increases, then decreases as n_1 grows. This occurs because the higher chance of selecting the optimal dose initially contributes to the higher power in the phase 3 trial. However, as n_1 grows, the reduced sample size in phase 3 offsets this benefit, leading to a decline in power. As the ORR of the optimal dose increases from scenario 1 to scenario 4, the probability of selecting it also increases for a given n_1. Consequently, the power consistently increases for both approaches, as a higher probability of selecting the optimal dose for the phase 3 portion at a given sample size directly boosts the power to reject the null hypothesis for the optimal dose. Note that in scenario 4, when the ORR of the optimal dose is high enough to ensure its selection into the phase 3 portion, increasing n_1 no longer enhances the power of the proposed design, as the probability of selecting the optimal dose is already close to 100%.

Between the two approaches, the Seamless Phase2/Phase3 offers slightly higher power than the Sequential Phase2/Phase3 when n_1 is small. The power advantage increases as n_1 grows. When n_1 is large, the power gain can reach up to 15%. These results suggest that the added complexity of conducting the adaptive design may not be worthwhile when the sample size planned for dose selection is relatively small (e.g., <10%) compared to the sample size for phase 3 hypothesis testing. However, when the sample size for dose selection is not insignificant relative to the phase 3 sample size, the adaptive design could provide a significant advantage.

We also conduct simulations to demonstrate the benefit of including an interim analysis in the phase 3 portion. In Table 7, the ORR remains the same across scenarios, while the mOS increases from 20.5 months to 22.5 months. For the proposed design, one interim analysis is planned at 388 OS events followed by a final analysis at 473 events. For comparison, a two-stage design is considered, with the final analysis (stage-2) in phase 3 at 473 events. Since ORR and n_1 are fixed, the probability of selecting the optimal dose remains at 92% across scenarios. The power for both approaches increases as the mOS increases. Given that the treatment effect in OS is assumed to be constant over time, the power is similar between the two approaches.

Table 6. Power comparison between seamless Phase2/Phase3 and sequential Phase2/Phase3.

			Power for the optimal dose	
Scenario	n_1	Probability of selecting the optimal dose	Seamless Phase2/ Phase3	Sequential Phase2/ Phase3
ORR: **0.25**, 0.2, 0.2, 0.1 mOS: 20.5, 18, 18, 15 $\rho = 0.4$				
	30	0.45	0.41	0.40
	60	0.54	0.48	0.45
	90	0.60	0.54	0.47
	120	0.66	0.59	0.49
ORR: **0.3**, 0.2, 0.2, 0.1 mOS: 20.5, 18, 18, 15 $\rho = 0.4$				
	30	0.66	0.59	0.57
	60	0.81	0.72	0.68
	90	0.88	0.78	0.69
	120	0.93	0.82	0.67
ORR: **0.35**, 0.2, 0.2, 0.1 mOS: 20.5, 18, 18, 15 $\rho = 0.5$				
	30	0.79	0.72	0.70
	60	0.92	0.82	0.78
	90	0.97	0.85	0.76
	120	0.99	0.87	0.71
ORR: **0.4**, 0.2, 0.2, 0.1 mOS: 20.5, 18, 18, 15 $\rho = 0.5$				
	30	0.91	0.82	0.80
	60	0.98	0.87	0.84
	90	0.99	0.87	0.79
	120	0.99	0.87	0.72

The interim or final analysis is event-driven and is expected to take longer to occur as the mOS increases because it slows downs event accrual. As a result, the expected study duration increases as the mOS increases. With similar power between the two approaches, adding one interim analysis significantly and consistently reduces the expected study duration across scenarios. In addition, the results in Table 7 also suggest that the power of the proposed design is bounded by the probability of selecting the optimal dose.

Four values in the first column represent the ORR and mOS in dose 1, dose 2, dose 3 and the control arm, respectively. Dose 1 is set as the optimal dose, with ORR increasing progressively from Scenario 1 to Scenario 4. n_1 is the sample size per arm at stage 1 for dose selection. Power of "Seamless Phase2/Phase3" represents the probability that the trial will reject null hypothesis for OS of dose 1, based on the proposed design. Power of "Sequential Phase2/Phase3" represents the probability that dose 1, will be selected from a phase 2 trial, and then will reject the null hypothesis for OS during either an interim analysis or in the final analysis of a separate group sequential phase 3 trial. Each row is based on 5000 simulations of a trial.

Four values in the first column represent the ORR and mOS in dose 1, dose 2, dose 3 and the control arm, respectively. Dose 1 is set as the optimal dose, with mOS increasing progressively

Table 7. Comparison of study duration between seamless Phase2/Phase3 with IA in phase 3 and seamless Phase2/Phase3 without IA in phase 3.

Scenario	Probability selecting the optimal dose	Seamless Ph2/Ph3 with IA in Ph3		Seamless Ph2/Ph3 without IA in Ph3	
		Power	Average study duration (mos)	Power	Average study duration (mos)
ORR: 0.35, 0.2, 0.2, 0.1 mOS: **20.5**, 18, 18, 15 $\rho = 0.5$, $n_1 = 60$	0.92	0.82	36.3	0.82	50.0
ORR: 0.35, 0.2, 0.2, 0.1 mOS: **21**, 18, 18, 15 $\rho = 0.5$, $n_1 = 60$	0.92	0.86	37.3	0.86	50.5
ORR: 0.35, 0.2, 0.2, 0.1 mOS: **21.5**, 18, 18, 15 $\rho = 0.5$, $n_1 = 60$	0.92	0.88	37.5	0.88	51.0
ORR: 0.35, 0.2, 0.2, 0.1 mOS: **22**, 18, 18, 15 $\rho = 0.5$, $n_1 = 60$	0.92	0.90	37.8	0.90	51.5
ORR: 0.35, 0.2, 0.2, 0.1 mOS: **22.5**, 18, 18, 15 $\rho = 0.5$, $n_1 = 60$	0.92	0.92	38.1	0.92	52.0

from Scenario 1 to Scenario 5. "Seamless Ph2/Ph3 with IA in Ph3" refers to the proposed design that includes one interim analysis at 388 OS events in the phase 3 portion followed by a final analysis at 473 OS events. "Seamless Ph2/Ph3 without IA in Ph3" refers to a design with only one final analysis in the phase 3 portion at 473 OS events. Study duration is defined as the time from first patient randomized to the first rejection of the OS null hypothesis. Each row is based on 5000 simulations of a trial.

5. Discussion

The proposed adaptive design uses combination tests for statistical testing. It is crucial to pre-specify the weights to ensure the control of Type I error rate. In this article, we set the weights to be same for all possible hypotheses. Different weights may be pre-specified for each hypothesis if the expected number of events differs among dose arms. For time-to-event endpoint, it is also crucial to base the two components of the combination test on independent cohorts of patients enrolled before and after dose selection. This ensures that no additional follow-up from the first cohort of patients contributes to the second component of the combination test, preventing any bias in its null distribution.

The proposed design uses a surrogate endpoint to predict the treatment benefit in long-term endpoint tested in the phase 3 portion. It is important to choose a surrogate endpoint for which the treatment effect is predictive of the treatment effect in the primary endpoint. In addition, as demonstrated in the simulation, the power of the study depends on the probability of selecting the optimal dose based on a surrogate early endpoint. It is critical to ensure there is adequate information to select the optimal dose at the interim analysis. If there is a concern that the benefit presented in the surrogate endpoint may not translate to the long-term endpoint, a futility analysis may be added to the phase 3 portion to mitigate the risk of exposing patients to a new drug that lacks long-term benefit.

The proposed design requires careful study monitoring. We recommend pausing enrollment at stage 1 for dose selection. Timely decision-making for dose selection is important to ensure smooth transition to the phase 3 portion. Alternatively, enrollment can remain open during stage 1 to maintain momentum. The downside of this approach is that more patients will be enrolled to the unselected dose arms. The choice of the approach will depend on the enrollment rates and other operational considerations on a case-by-case basis. In addition, survival follow-up will be required for the unselected dose arms. Their OS data will be included in the testing of the intersection hypotheses involving the selected dose arm to adjust for comparisons of multiple dose arms against a common control arm. While this may appear to add operational burden, patients in oncology trials often experience disease progression relatively soon after start of treatment, leading to discontinuation of treatment. Following up for OS typically requires only phone call visits, which does not impose significant operational demands. In addition, assessing their long-term survival data is important to support the selected dose, particularly if the selected dose leads to a regulatory submission.

Lastly, we want to emphasize the importance of the careful statistical considerations for the proposed adaptive design. As shown in the simulation, the proposed adaptive design may not always offer a significant advantage to justify the added complexity compared to the traditional sequential phase 2 followed by phase 3 approach. The power gain provided by the proposed design, compared to the traditional sequential Phase 2/Phase 3 approach, depends on the efficacy settings of the early surrogate and primary endpoints, as well as the sample sizes used for dose selection and subsequent Phase 3 hypothesis testing. Based on our experience by simulations, the power gain is typically limited when the sample size planned for dose selection is less than 10% of the sample size allocated for phase 3 hypothesis testing. We recommend conducting simulations to evaluate the operating characteristics and ensure the desired advantages are achieved under the specific settings. The sample size and timing for dose selection need to ensure sufficient information is available to select the optimal dose. Likewise, the sample size and timing for interim analyses in the phase 3 portion need to provide adequate information to detect the expected treatment effect while accounting for multiplicity adjustment. To facilitate the use of the proposed design, we have developed an R package available at: *https://github.com/eric-zhang16/gsDose*.

Disclosure Statement

No potential competing interest was reported by the authors.

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